Local Stability and Topological Structure of an Adaptive DeFi Risk Controller

A 360-point parameter-neighbourhood experiment

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Scope. This report tests whether convergent controller coefficients form a robust local neighbourhood and whether the sampled stable set has a detectable topological structure. It combines a reduced-order closed-loop stability probe, 138,240 four-month economic paths, and persistent homology with a matched random-cloud null test. It is computational evidence, not a formal Lyapunov proof or a production deployment claim.

Experiment seed: `20261105`

Repository:
https://github.com/nikorokni/local-currency-defi-adaptive-governance

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1 Executive summary

The published controller uses fixed coefficients in a bounded, rate-limited map from observed volatility, DEX basis, oracle disagreement, and utilisation to the borrowing rate, debt ceiling, liquidation ratio, and circuit-breaker state. The original simulations establish economic performance under common shocks, but they do not show whether one successful coefficient vector is surrounded by other convergent vectors.

This report provides that missing computational experiment. A 96-point broad Latin-hypercube pilot located a stable centre. Exactly 360 stratified random points were then sampled around that centre. Each point was tested from nine perturbed initial states in six stability cases (three stress regimes, each at zero and 48-hour delay), and was separately run on 192 common ARS paths and 192 common TRY paths.

Headline result	Value
Broad pilot points robustly convergent	11 / 96
Local points robustly convergent	**360 / 360**
Local points converging in all six cases	261 / 360
Four-month economic paths in local stage	138,240
Median pooled bad-debt-event probability	0.2604%
Local stable-set components at connection scale	1
Observed maximum H_1 persistence	0.1844
Matched-null p-value for maximum H_1 persistence	0.9355
Evidence of non-trivial one-dimensional hole	**No**

The evidence supports a connected local stability basin. It does not support a robust ring or hole after comparison with uniform eight-dimensional Latin-hypercube clouds. Stability is not identical to economic safety: TRY severe depegs remain in the common-path experiment even when controller states converge.

2 Controller and parameter space

2.1 Parameterised equations

The experiment preserves the published baseline by exposing eight coefficients that had previously appeared as literals. The EWMA variance is

$$v_{t+1} = \beta v_t + (1-\beta) q_t^2, \tag{1}$$

and the composite risk score is

$$R_t = \mathrm{clip}_{[0,1]}\left(w_\sigma \sqrt{v_{t+1}} + w_b b_t + w_\delta \delta_t + w_u (u_t - 0.72)_+\right). \tag{2}$$

The three parameter targets are

$$r_t^* = \mathrm{clip}_{[0.04,1.50]}\left(0.08 + g_r R_t + 0.35(u_t - 0.75)_+\right), \tag{3}$$

$$C_t^* = \mathrm{clip}_{[0.20,1.00]}\left(e^{-g_C R_t}\right), \tag{4}$$

$$\lambda_t^* = \mathrm{clip}_{[1.50,2.20]}\left(1.50 + g_\lambda R_t\right). \tag{5}$$

The published per-step movement bounds remain unchanged. Therefore setting θ to the baseline row in Table 1 reproduces the original equations exactly.

2.2 Parameter collection

Table 1: Influential controller coefficients and broad search intervals.

Coefficient	Symbol	Baseline	Broad interval
EWMA memory	β	0.92	$[0.80, 0.98]$
Volatility weight	w_σ	4.00	$[2.50, 5.50]$
DEX-basis weight	w_b	3.20	$[2.00, 4.50]$
Disagreement weight	w_δ	1.80	$[1.00, 2.80]$
Utilisation weight	w_u	0.90	$[0.40, 1.40]$
Rate response gain	g_r	0.90	$[0.50, 1.30]$
Ceiling response gain	g_C	1.75	$[0.80, 2.80]$
Liquidation response gain	g_λ	0.70	$[0.35, 1.10]$

The selected pilot centre was

$$\hat{\theta} = (0.94569, 3.17732, 2.31667, 1.48870, 0.70887, 0.52487, 0.81518, 0.57902). \tag{6}$$

It is a centre for local robustness sampling, not a claimed economic optimum.

3 Experimental design

3.1 Meaningful random sampling

Independent uniform draws can cluster and leave large gaps. We instead use a deterministic Latin-hypercube design: every parameter marginal is divided into equal-probability strata and each stratum is visited once. A 96-point pilot covers the broad box. A second 360-point design covers a neighbourhood equal to 18% of each broad-range width on either side of the stable centre, clipped to the broad bounds. All randomisation is controlled by seed `20261105`.

3.2 Reduced closed-loop stability probe

The stability state is

$$z_t = (r_t, C_t, \lambda_t, D_t, b_t, v_t), \tag{7}$$

where demand responds negatively to the rate and available ceiling, debt accrues interest and is repaid, DEX basis responds to accepted draws and arbitrage capacity, and the breaker can pause new draws. This reduced model preserves the feedback signs, hard domains, step bounds, and breaker asymmetry of the full simulator.

Nine perturbed initial states are evaluated in calm, moderate, and crisis regimes. Each regime is run both without delay and with an eight-step (48-hour) delay. A case passes only if all state variables remain finite and in-domain and if

$$\overline{\|z_t - z_{t-1}\|}_{\text{tail}} < 0.002, \tag{8}$$

$$\text{range}_{\text{tail}}(z) < 0.030, \tag{9}$$

$$\text{spread}(z_T) < 0.040, \tag{10}$$

$$\text{spread}(z_T)/\,\text{spread}(z_0) < 0.50. \tag{11}$$

A vector is robustly convergent when all six cases remain bounded, at least five cases pass, and the worst terminal spread is below 0.055.

3.3 Full economic cross-check

Every local point is evaluated with the full lending-book simulator on 192 ARS and 192 TRY four-month paths, each containing 480 six-hour steps. The same market paths are reused for every point. This common-random-number design removes shock-sampling variation from comparisons between parameter vectors.

4 Stability results

Only 11 of the 96 broad pilot points passed the robust criterion. The published baseline remained bounded but converged in four of six reduced cases; it failed the stricter cross-perturbation contraction requirement. This should not be read as proof that the published finite-horizon economic simulation is unstable. Rather, it identifies a gap between finite-path performance and local-attractor evidence under the additional frozen-regime tests.

All 360 local points passed the robust criterion. Of these, 261 passed all six cases and 99 passed five. The local cloud therefore supplies evidence that the selected convergent values are not isolated.

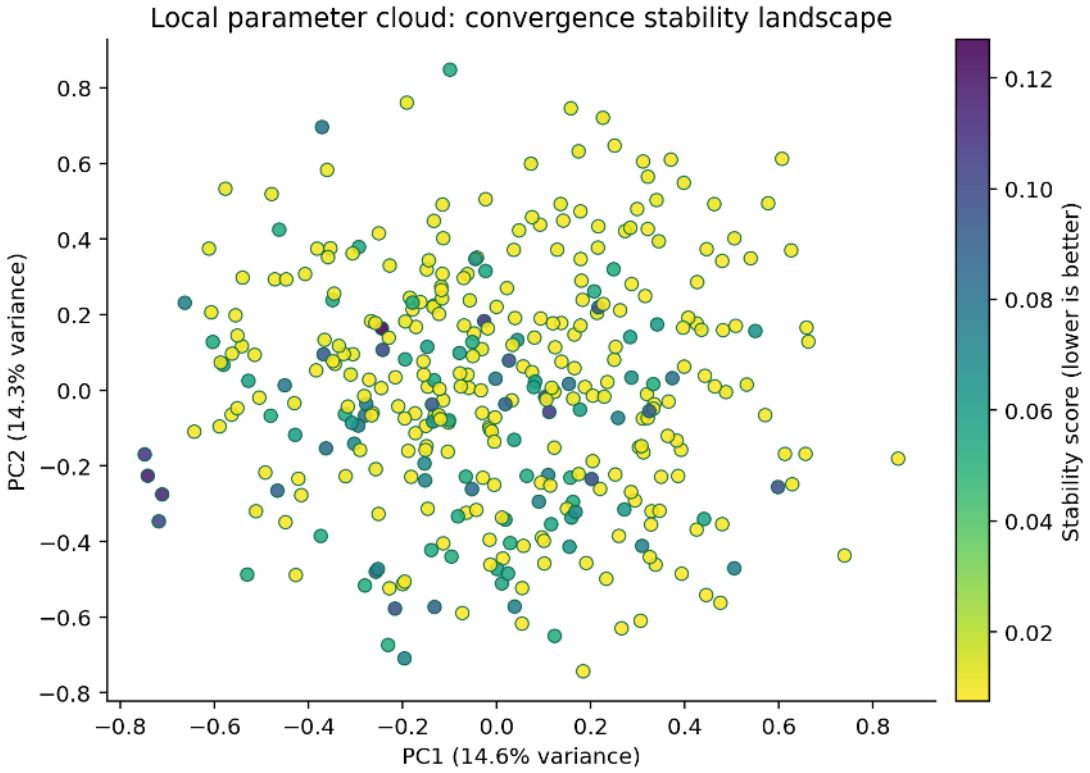


Figure 1: PCA view of the eight-dimensional local cloud. Edge colour denotes the robust label; fill colour is the continuous stability score. The first two PCs explain only 28.9% of parameter variance, so this plot is illustrative rather than a complete representation of geometry.

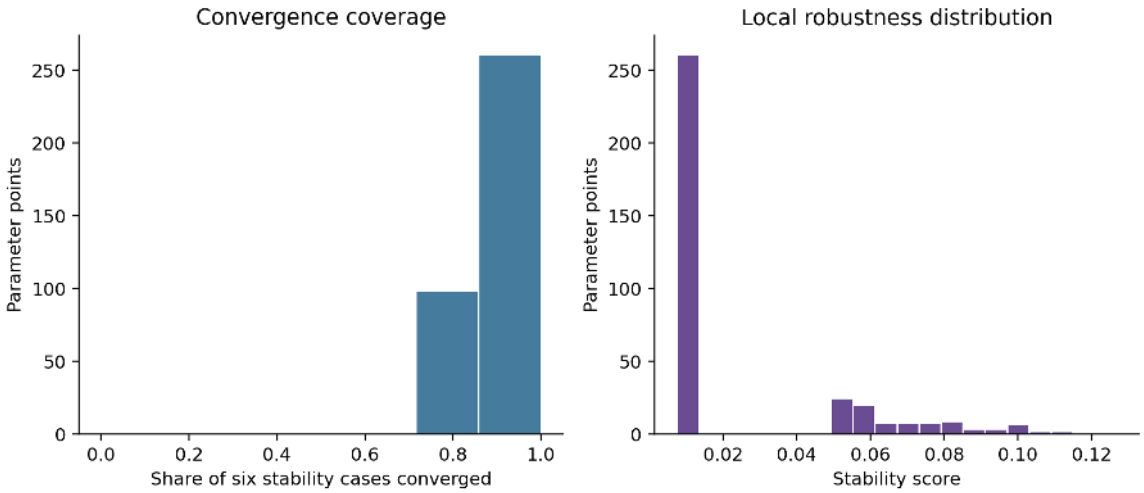


Figure 2: Coverage of the six convergence cases and distribution of the continuous stability score. Lower scores indicate smaller tail movement, spread, and contraction penalties.

5 Economic results

Across the 360 local configurations, pooled bad-debt-event probability ranged from 0 to 1.3021%, with median 0.2604%. Mean issued credit ranged from 0.1948 to 0.2307 and pause share from 0.4092 to 0.6184. The median pooled credit was 0.2147.

ARS bad-debt probability averaged 0.2951%, while TRY averaged 0.3935%. ARS had no severe-depeg event in this particular common set. TRY severe-depeg probability stayed close to 20.8%. Thus, robust convergence of the policy state does not imply that all exogenous or liquidity-driven failure modes disappear.



Figure 3: Credit, bad-debt-event probability, and breaker use. Each marker is one parameter vector evaluated on the same ARS and TRY paths.

5.1 Parameter influence

Spearman correlations are used as descriptive local sensitivity measures. They are not structural causal effects and their nominal p-values are not used as formal model-selection tests.

- Ceiling gain versus stability score: $\rho = 0.931$. Within this local cloud, stronger ceiling response is associated with weaker convergence metrics.
- Liquidation gain versus bad-debt probability: $\rho = -0.839$.
- Rate gain versus pause share: $\rho = -0.587$.
- Ceiling gain versus mean credit: $\rho = -0.550$.

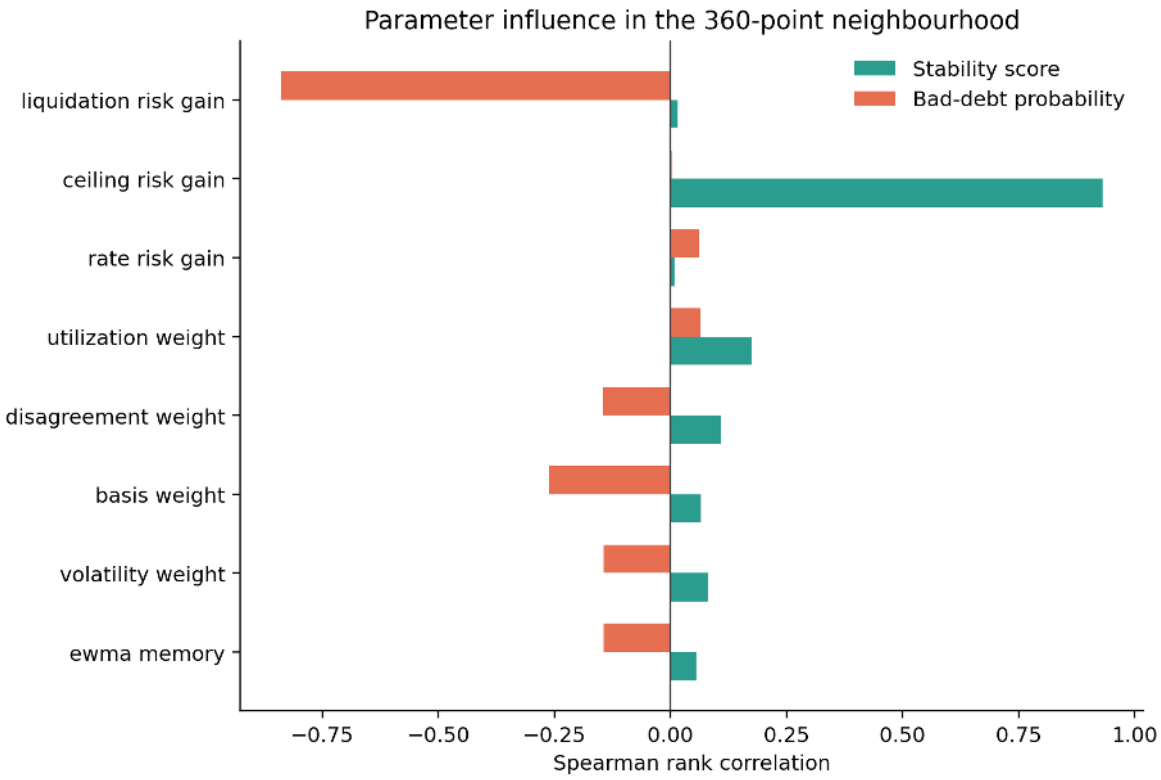


Figure 4: Local rank correlations with the stability score and bad-debt-event probability. Signs describe association in the sampled neighbourhood only.

6 Topology of the convergent region

The eight parameter coordinates are mapped to the unit cube before computing a Vietoris–Rips filtration. The number of connected components is the zeroth Betti number β_0, while β_1 counts one-dimensional cycles at a given filtration radius.

The stable cloud becomes connected at radius 0.7390. Its largest finite H_1 lifetime is 0.1844. High-dimensional finite point clouds commonly produce transient cycles even when sampled from a filled box, so the existence of cycles alone is not evidence of a genuine hole. We therefore repeat the maximum-persistence statistic on 30 matched eight-dimensional Latin-hypercube null clouds.

The null median is 0.1980 and the null 95th percentile is 0.2265. The empirical tail probability for a null maximum at least as large as the observed value is $p = 0.9355$. Consequently, the apparent cycles are fully compatible with finite sampling from a filled region.

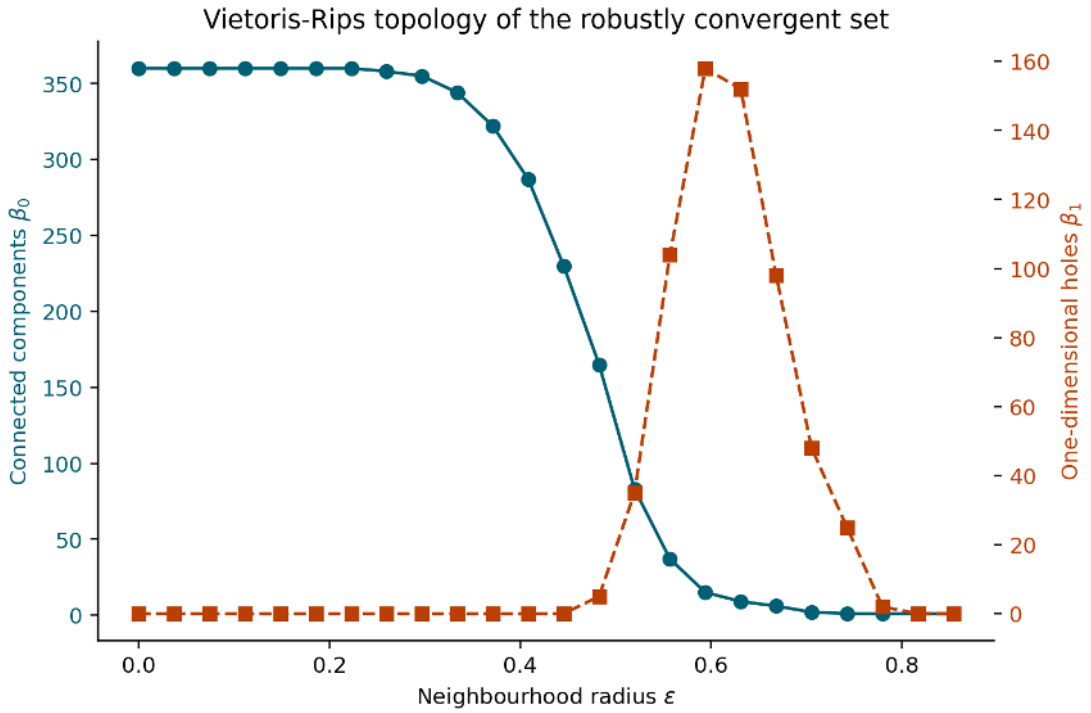


Figure 5: Betti numbers over filtration radius. The point cloud becomes one component and all sampled H_1 cycles eventually die. The matched null test, not visual persistence alone, determines whether the cycles are exceptional.

The defensible topological conclusion is therefore:

> The sampled convergent set contains a connected local basin. There is no evidence of a non-trivial one-dimensional hole beyond the topology generated by uniform high-dimensional sampling.

This is not a proof that the global stable set is connected or contractible.

7 What the experiment establishes

- **Local robustness:** convergence is retained throughout 360 stratified perturbations around the selected centre.
- **Non-isolation:** the stable point is part of a measurable local neighbourhood rather than a single tuned vector.
- **Connected sample geometry:** the local stable cloud joins into one component at a finite filtration scale.
- **No exceptional hole:** observed H_1 persistence is smaller than typical maxima from matched filled-box null clouds.
- **Separate safety claim:** convergence and economic outcomes remain distinct; TRY depeg risk persists in some common paths.

8 Limitations and next proof obligations

1. The study is empirical and finite. A Lyapunov or Lyapunov–Krasovskii argument is still required for a mathematical convergence theorem, particularly under delay and boundary projection.

2. The reduced stability model preserves signs and bounds but is not identical to every stochastic mechanism in the full lending simulator.

3. The topology is local and sample-dependent. The broad parameter box may contain other stable components that the pilot did not resolve.

4. Stability thresholds are explicit design choices. Future work should vary them and report label sensitivity.

5. Rank correlations can be affected by nonlinear interaction and should not be read as causal derivatives.

6. A stable equilibrium can still be economically undesirable. Safety, credit availability, liquidation cost, and depeg resilience require joint multi-objective evaluation.

9 Reproducibility map

Repository path	Purpose
`analysis/stability_topology.py`	Sampling, reduced-loop stability, Betti numbers, persistent homology, and influence measures.
`analysis/run_stability_topology.py`	End-to-end 96+360 point experiment, full economic runs, CSVs, figures, hashes, and summary.
`results/stability_topology/pilot_parameter_search.csv`	Broad 96-point search and stability metrics.
`results/stability_topology/local_360_parameter_results.csv`	All local parameters, convergence metrics, and pooled economic outcomes.
`results/stability_topology/economic_results_by_currency.csv`	720 point–currency records.
`results/stability_topology/topology_persistence_intervals.csv`	Full zero- and one-dimensional persistence intervals.
`results/stability_topology/topology_betti_curves.csv`	Betti numbers across filtration radius.
`results/stability_topology/parameter_influence.csv`	Spearman correlations and nominal p-values.
`results/stability_topology/experiment_summary.json`	Seed, bounds, headline results, topology null test, and SHA-256 output hashes.
`tests/test_stability_topology.py`	Equation-preservation, sampler, boundedness, and topology unit tests.

Conclusion

The 360-point experiment answers the motivating question affirmatively but with an important qualification. Around the pilot-selected centre, convergence is stable to simultaneous perturbations of all eight influential coefficients. The sampled region is connected, and its transient cycles do not exceed those expected from a uniformly filled eight-dimensional cloud. The result is robust computational evidence for a local stability basin; it is not yet a global topological theorem or a formal proof of delayed closed-loop convergence.